

Teaching Programming at a Distance

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ABSTRACT

Novice programmers face significant barriers when they cannot learn face-to-face with a teacher or with peer support in a physical computer laboratory. Online collaborative programming environments (virtual computing laboratories) enable an overview of multiple students programming at the same time, with tutors able to interact with and support individual students or groups in real-time.

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1 BACKGROUND / RELATED WORK

It has been recognised that collaborative learning platforms [1] could be an effective aid for novice programmers. Indeed, Carter et al. [2] suggest that social factors heavily contribute to learners' eventual success or failure, with Xinogalos et al. [3] finding that collaborative problem solving helps students to be more confident.

Furthermore, Maguire et al. [4] investigated a range of factors with students starting introductory programming classes, including sociodemographic factors, engagement, and social media usage. They found that confidence in programming is an independent predictor of exam performance when learning to program.

1.1 Research Goals

Early research goals have been established with a pilot research project. The aim of the project is to investigate and produce recommendations for the use of virtual computer laboratories, when teaching programming at a distance.

1.2 Research Question

RQ1: Does the use of a virtual computer laboratory environment during synchronous tutorials improve the teaching and learning of programming at a distance?

1.3 Method

The pilot research project is currently in progress and the following describes the method used.

At the start of a first-year computing degree course (April 2022), students were invited to participate in the research project. Students

were provided with a survey which captured prior programming experience and confidence in programming. Confidence in programming was assessed using the Computing Attitudes Survey [5] which is a validated instrument.

Students were subsequently provided with access to a virtual computer laboratory environment (using either Coding Rooms or REPLIT) during six practical programming sessions. Student performance in programming tasks was observed during sessions.

Tutor-student and student-student interactions during these practical programming sessions were also observed to evaluate teaching strategies used when teaching programming at a distance, specifically relating to social interaction, technology utilisation and collaboration.

This has provided some insight into the impact of different strategies, over time, in relation to confidence, student outcomes and perceptions of online collaborative environments.

At the end of the 6-month module, students will complete the Computing Attitudes Survey again, and will provide comments on their perceptions of using virtual computing lab environments. Feedback from tutors will also be gathered during a focus group.

Once the module results are in, data relating to student performance in their assignments, specifically in programming questions, will be analysed. Following analysis of all the data, initial recommendations will be made for using virtual computing lab environments.

The surveys, focus group and module results will provide quantitative as well as qualitative data in relation to the impact that this intervention has had, as well as perceptions into the use of virtual computing laboratories.

This research is expected to contribute to approaches to teaching programming where students and teachers are not collocated.

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